DumpAnalyzer

z/OS SYSDUMP & CEEDUMP Instant Analysis Tool

https://dumpanalyzer-production.up.railway.app

Upload any IBM z/OS SYSDUMP or CEEDUMP file and receive instant root-cause analysis, z/Architecture instruction decoding, PSW arithmetic, COBOL fix recommendations, and integrity flag detection.

IBM z/OS | COBOL & Language Environment | S0C7 · S0C4 · S0CB · S322 · S806 | Node.js · Express · Railway

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GETTING STARTED

How to Use DumpAnalyzer

DumpAnalyzer is a zero-install web tool. Open the URL in any browser, upload your dump file, and get analysis in seconds. No account or login required.

Step-by-Step Guide

1. **Open the tool in your browser**

   Navigate to **https://dumpanalyzer-production.up.railway.app**. You will see the DumpAnalyzer home screen with the upload area and feature summary.

2. **Upload your SYSDUMP or CEEDUMP file**

   Click **Choose File** or **Upload Dump** (top right), or drag & drop your dump file directly onto the page. Accepted extensions: `.dmp` `.txt` `.log` `.cee` `.sys` `.sdmp`. Maximum file size: **50 MB**.

3. **Review the analysis output**

   The tool automatically parses and analyzes the dump. The full-screen output contains: abend summary, root cause box, PSW arithmetic, instruction decode, fix recommendations, and integrity flags.

4. **View the raw dump text (optional)**

   Click the **View Raw** button (top right of the analysis page) to open a modal with the original dump text for cross-reference. Press `Escape` to close it.

5. **Analyze another dump**

   Click ← **New Analysis** (top left) to return to the home screen and upload a different file.

Supported Dump Formats

SYSDUMP (IEA992I / IEA995I) — Internally generated by z/OS when a job abends. Contains PSW, registers, storage.

SYSDUMP (real z/OS format) — Files from actual z/OS jobs with `1JOB` header, `COMPLETION CODE SYSTEM =`, and embedded `CEE3207S` diagnostics.

CEEDUMP (CEE3DMP) — Language Environment structured dump with traceback, working storage, and linkage section.

UNDERSTANDING THE OUTPUT

Analysis Output Sections

After uploading, the output is organized into these sections from top to bottom:

Summary Grid

Six key metrics at a glance: **Abend Code** (e.g. S0C7), **Program Name**, **Dump Type** (SYSDUMP/CEEDUMP), **Statement Number**, **Offset** (hex), and **Timestamp**. For SYSDUMPs, also shows Job Name and Step Name.

Root Cause Analysis Box

The most important section. Shows the abend classification, the failing field and its bad value (e.g. `IN-POL-VALUE = X'404040'`), the COBOL statement, and a full narrative explanation of why the abend occurred and what it means technically.

PSW Arithmetic — Failing Instruction Location

Decodes the Program Status Word to calculate the exact address of the failing instruction: **Failing Instruction Address = PSW Word 2 − ILC**. Shows the next instruction address, the calculated failing instruction address, and the module offset.

Machine Instruction at Abend

Decodes the raw hex bytes at the failing address into a z/Architecture instruction. For SS-format instructions (AP, SP, MP, DP, ZAP etc.) shows opcode, length fields, base register + displacement for both operands, and the full decode detail.

Abend Technical Detail

INTCODE, trigger instructions, and COBOL data types affected by this abend type. Full technical description of the hardware exception.

Fix Recommendations

Three-part fix guidance: (1) numbered restart steps with FILE-AID / SORT utility commands, (2) ready-to-paste COBOL code fix with IS NUMERIC guard, (3) upstream ETL/extract recommendation to prevent recurrence.

Dump Integrity Flags

Automatically detected inconsistencies — e.g. S322 found in a CEEDUMP (unusual), ILC ≠ 6 for a packed-decimal abend, 24-bit AMODE when 31-bit expected.

REFERENCE

Supported Abend Codes & API

Recognized z/OS Abend Codes

ABEND CODE	NAME	COMMON CAUSE IN COBOL
S0C7	Data Exception	COMP-3 field contains EBCDIC spaces (X'40') or invalid nibbles — triggers on AP/SP/MP/DP/ZAP/CP/ED
S0C4	Protection Exception	Null or uninitialized COBOL POINTER used as base register — triggers on MVC/ST/L
S0CB	Decimal Divide Exception	DIVIDE statement or COMPUTE with / operator where divisor is zero — triggers on DP
S322	CPU Time Limit Exceeded	Infinite PERFORM UNTIL loop (exit condition never set) — job cancelled by JES2 timer
S806	Load Module Not Found	CALL to a sub-program whose load module is not in STEPLIB, JOBLIB, or LINKLIST

REST API Endpoint

The analyzer exposes a single endpoint for programmatic use:

POST /api/analyze

Send a `multipart/form-data` request with field `dumpFile` containing the dump file. Returns a JSON object with `type`, `summary`, parsed sections, and a rich `analysis` object including `rootNarrative`, `instrDecode`, `pswDecode`, `cobolFix`, `immediateSteps`, and `integrityFlags`.

Data Privacy

DumpAnalyzer processes files entirely in memory — uploaded dump files are **not persisted to disk** on the server. Each request is stateless. Do not upload dumps containing production customer PII without appropriate data classification approval.

Links

https://dumpanalyzer-production.up.railway.app

GitHub: **https://github.com/anuragbhumca07/DumpAnalyzer**

DumpAnalyzer · Built with Node.js, Express, Playwright · Deployed on Railway · September 2026